

PRACTICAL WEEK-3
2520090210
KODAM SATHWIK

The image displays two screenshots of the MongoDB Data Explorer interface, showing the 'book_metadata' collection in the 'bookflow' database. The interface includes a sidebar with 'CLUSTERS (1)' and a main panel for document viewing.

Top Screenshot: Shows a single document with the following details:

- Document 1:**
 - `_id`: ObjectId('6a91432f6d442b6cd5cf3e88')
 - `book_id`: 101
 - `title`: "The Martian"
 - `author`: "Andy Weir"
 - `published_year`: 2021
 - `category`: "Science Fiction"
 - `format`: "Digital PDF"
 - `file_size`: "2.5 MB"
 - `member_id`: 101
 - `reviews`: Array (3)

Bottom Screenshot: Shows a list of documents, with the first document expanded to show details:

- Document 1:**
 - `_id`: ObjectId('6a9143ae6d442b6cd5cf3e86')
 - `book_id`: 105
 - `title`: "Data Structures"
 - `author`: "Mark Allen"
 - `published_year`: 2024
 - `category`: "Computer Science"
 - `format`: "Digital PDF"
 - `file_size`: "5.0 MB"
 - `member_id`: 105
 - `reviews`: Array (2)

Both screenshots show the 'Alert System Status: All Good' and copyright information at the bottom.

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

local

studentDB

students

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

{ "reviews.rating": { "\$gt": 4 } }

Explain Reset Find Options

member_id: 101
rating: 5
comments: "Excellent science fiction story"

1: Object (3)
member_id: 102
rating: 4
comments: "Very interesting and informative"

2: Object (3)
member_id: 103
rating: 5
comments: "Excellent space adventure"

Alert System Status: All Good

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Documents 5 Aggregations Schema Indexes 1 Validation

{ "reviews.rating": { "\$in": [4, 5] } }

Explain Reset Find Options

0: Object (3)
member_id: 101
rating: 5
comments: "Excellent science fiction story"

1: Object (3)
member_id: 102
rating: 4
comments: "Very interesting and informative"

2: Object (3)
member_id: 103
rating: 5
comments: "Excellent space adventure"

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Documents 5

Aggregations

Schema

Indexes 1

Validation

{

"format": {

"\$regex": "Digital",

"\$options": "i"

}

}

Explain

Reset

Find

Options

25

1-4 of 4

<

>

⋮

{}

⌕

_id: ObjectId('6a91432f6d442b6cd5cf3e88')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

101

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CLUSTERS (1)

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bookflow

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local

studentDB

students

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

\$match

Explain

Run

Options

Untitled - modified

+ CREATE NEW

PREVIEW

{}

</>

⚙

Stage 1 | \$match

1 {

2 "published_year": {

3 "\$gt": 2020

4 }

5 }

Output preview after \$match stage (Sample of 4 documents)

_id: ObjectId('6a91432f6d442b6cd5cf3e88')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

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Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

Explain

Run

Options

Untitled - modified

Stage 2 \$unwind

```
1 /**
2  * path: Path to the array field.
3  * includeArrayIndex: Optional name for
4  * preserveNullAndEmptyArrays: Optional
5  * toggle to unwind null and empty val
6  */
7 {
8   path: "$reviews"
9 }
10
```

Output preview after \$unwind stage (Sample of 10 documents)

```
_id: ObjectId('6a9143ae6d442b6cd5cf3e86')
book_id: 185
title: "Data Structures"
author: "Mark Allen"
published_year: 2024
category: "Computer Science"
format: "Digital PDF"
file_size: "5.0 MB"
member_id: 185
```

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Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

\$group

Explain

Run

Options

Untitled - modified

Stage 3 \$group

```
1 {
2   "_id": "$title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
```

Output preview after \$group stage (Sample of 4 documents)

```
_id: "Data Structures"
average_rating: 4
total_reviews: 2
```

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Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

\$match

\$unwind

\$group

\$sort

+

?

Explain

Run

Options

Untitled - modified

+ CREATE NEW

</>

PREVIEW

{}

</>

⚙

Stage 4 \$sort

⌵

1 |

2 {

3 "average_rating": -1

4 }

Output preview after \$sort stage (Sample of 4 documents)

_id: "Dune"

average_rating: 4.666666666666667

total_reviews: 3

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Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

Create Index

Refresh

Looking for search indexes?

These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	4 (since Fri Aug 28 2026)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 kB	0 (since Fri Aug 28 2026)		READY

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CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
 - local
- studentDB
 - students

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 2Validation

{
 "\$text": {
 "\$search": "excellent"
 }
}

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_id: ObjectId('6a91432f6d442b6cd5cf3e80')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)

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ORGANIZATIONPROJECTIXSCAN1/1

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree

Raw Output

"parsedTextQuery": {
 "terms": ["excel"],
 "negatedTerms": [],
 "phrases": [],
 "negatedPhrases": []
},
"textIndexVersion": 3,
"inputStage": {
 "stage": "FETCH",
 "inputStage": {
 "stage": "IXSCAN",
 "keyPattern": {
 "_fts": "text",
 "_ftsx": 1
 },
 "indexName": "reviews.comments_text",
 "isMultiKey": true,
 "isUnique": false,
 "isSparse": false,
 "isPartial": false,
 "indexVersion": 2,
 "key": {}
 }
}

Query Performance Summary

5 documents returned

5 documents examined

0 ms execution time

Is not sorted in memory

5 index keys examined

Query used the following index:
_fts (text)
_ftsx ↑

Close

